LAB 5

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Exercise 4.3 Page 59

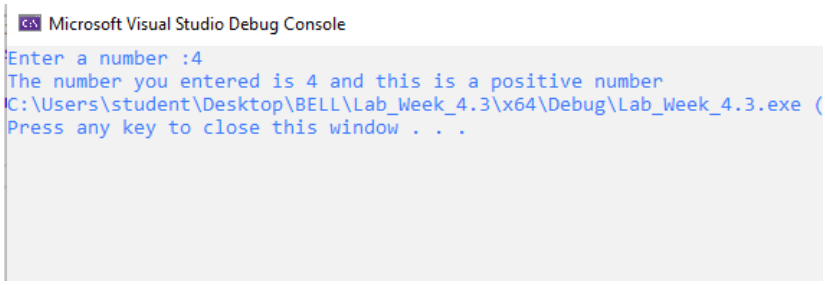
Question 1

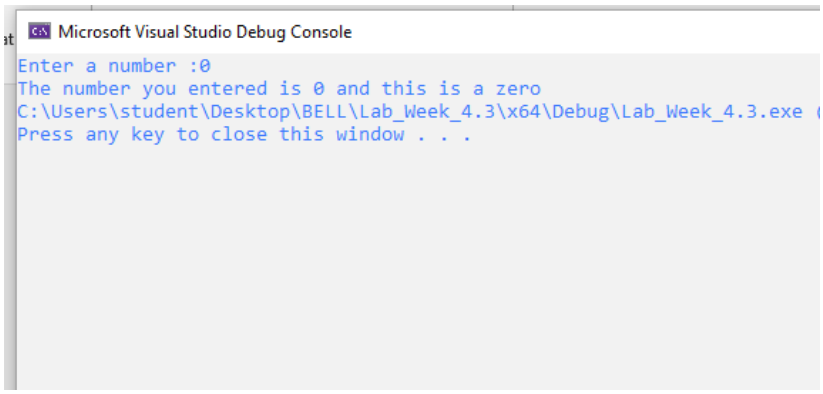
```
#include <iostream>
using namespace std;

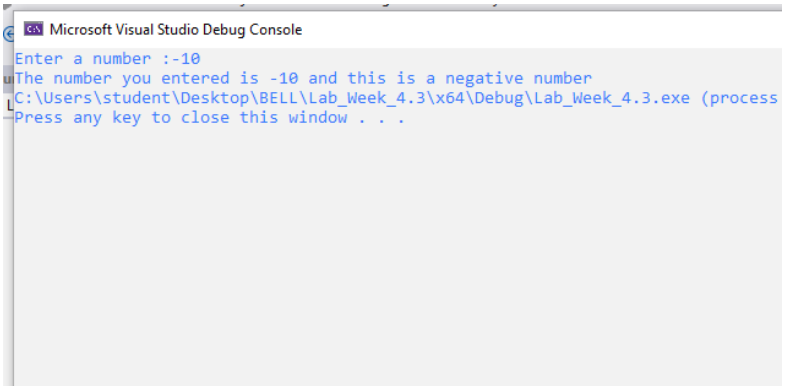
int main()
{
	int num; // declare
	cout << "Enter a number :";
	cin >> num;

	if (num > 0) {
		cout << "The number you entered is " << num << " and this is a
positive number";
	}
	else if (num == 0) {
		cout << "The number you entered is " << num << " and this is a
zero";
	}
	else {
		cout << "The number you entered is " << num << " and this is a
negative number";
	}
	return 0;
}
```

Output :







Question 2

```
#include <iostream>
#include <iomanip> // to set precision decimal place
using namespace std;

int main() {

	double current_cost, inflation_rate, rate; // declare
	int years; // declare

	cout << "Enter the current cost of the item : RM ";
	cin >> current_cost;

	cout << "Enter the current inflation rate : ";
	cin >> inflation_rate;

	cout << "Enter the number of years to estimate : ";
	cin >> years;
	cout << endl;

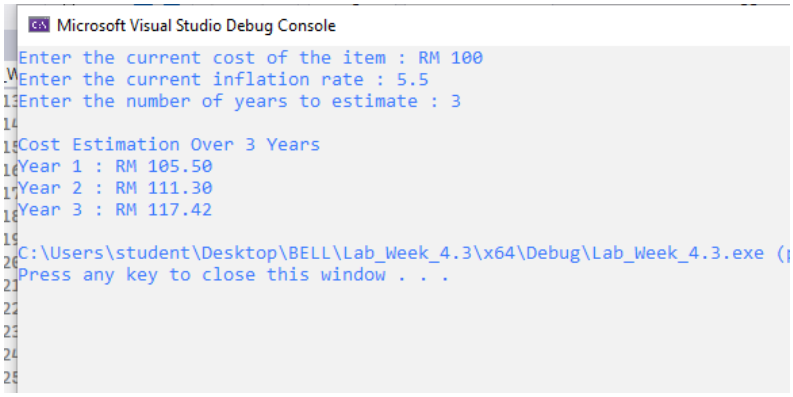
	if (current_cost < 0 || years <= 0) {
		cout << "Cost cannot be negative and years cannot be 0 or negative."
<< endl;
	}
	else {
		rate = inflation_rate / 100.0;
		double estimate_cost = current_cost;

		cout << fixed << setprecision(2);
		cout << "Cost Estimation Over " << years << " Years " << endl;

		for (int i = 1; i <= years; i++)
		{
			estimate_cost = estimate_cost + estimate_cost * rate;
			cout << "Year " << i << " : RM " << estimate_cost << endl;
		}
	}

	return 0;
}
```

Output :



Question 3

```cpp
#include <iostream>
using namespace std;

int main() {
    cout << "Prime numbers between 3 and 100:" << endl;

    for (int n = 3; n <= 100; ++n) {
        bool is_prime = true;

        for (int i = 2; i < n; ++i) {
            if (n % i == 0) {
                is_prime = false;
                break;
            }
        }

        if (is_prime) {
            cout << n << " ";
        }
    }
    cout << endl;

    return 0;
}
```

Output :

